

CEU Sign Up Sheet

Topic: **Oak Cynipidae, Pine Brown Spot, Needle Blight**

Date: **8/2/24**

Meeting length: 60 Minutes

Meeting Summary:

Meeting topics covered tree care and pest management strategies, including for trees beneath structures like decks. **Fertilization and treatment options** include **trunk injections** (e.g., MN Jet) and **soil drenches** for macro nutrients, with attention to **runoff control**. For **Dutch Elm Disease (DED)**, higher injections may help trees resist the disease. **Oak Bullet Gall** (*Disholcaspis quercusmamma*) and **Jumping Oak Gall** (*Neuroterus saltatorius*) are caused by cynipid wasps, with no harmful effects on tree health; they're primarily cosmetic. For **Brown Spot Needle Blight**, fungicide treatment and environmental protections like sanitation and moisture control are recommended.

Name

Signature

CEU Sign Up Sheet

Topic: **Case Study: White Oak Snelling Ave N**

Date: **8/11/24**

Meeting length: 60 Minutes

Meeting Summary:

Reviewed pictures of a local **White Oak** tree, and discussed the primary symptoms and potential causes of stress. The tree shows signs of decline, possibly linked to **Trunk and Root Disease (TCB)**, **Armillaria root rot**, and **overwatering**. The **base of the tree** exhibited **mechanical damage**, likely from **weed whipper injury**, creating significant wound response tissue. There is **moisture accumulation** around the trunk, suggesting a **perpetual moisture problem** possibly due to an irrigation system, which may exacerbate fungal growth. **Fungal growth** on the trunk and visible decay, combined with a **lightning strike injury** and **previous wounding**, points to long-standing stress and susceptibility to pathogens. Although **Oak Wilt** is a concern, it's less likely for White Oaks to "flash fry" from this disease, and it may not be the primary issue. Given the advanced decay and larvae in the stem, **TCB** and **Armillaria** are more plausible diagnoses. The tree's stress is compounded by a history of **mechanical damage**, poor water management, and potential fungal infections.

Name

Signature

CEU Sign Up Sheet

Topic: **Hemlock Rust Mite / Dutch Elm Disease**

Date: **8/26/24**

Meeting length: 60 Minutes

Meeting Summary:

Discussed and reviewed photos of Hemlock rust mite damage and trees with symptoms of DED.

Hemlock Rust Mite (*Nalepella tsugifoliae*) is an eriophyid mite that feeds on the interior of hemlock needles, causing a rusty yellow/gray appearance due to shedding exoskeletons. While nearly microscopic, they can be lethal when combined with other stress factors. Mite populations thrive in cool weather and are controlled with miticide in spring or fall. New growth is largely unaffected, and damage can resemble fungal diseases like **Dothistroma**. Abiotic factors, such as sun exposure and poor soil conditions, should be considered when diagnosing tree health.

Dutch Elm Disease (DED) is a fungal disease spread by elm bark beetles that clogs vascular tissue, leading to tree death. Early infections can be managed with **propiconazole** treatments. A customer's concern about DED was addressed by explaining that infection likely started last year, given the tree's history of neglecting injections. Upon inspection, the tree showed signs of storm damage rather than DED, with wilting branches and cracking, suggesting the issue may be mechanical rather than fungal.

Name

Signature

CEU Sign Up Sheet

Topic: **Common Conifer Pathogens**

Date: **9/9/24**

Meeting length: 60 Minutes

Meeting Summary:

Fungal diseases on conifers, such as **Rhizosphaera Needle Cast**, **Diplodia Shoot Blight**, **Dothistroma Needle Blight**, and **Brown Spot Needle Blight**, are spread by rain, sprinklers, and airborne spores. Infected needles or shoots cannot be cured but can be protected through persistent treatment over 2-3 years. Treatment involves spraying 3 times as new needles or shoots emerge, offering protection during their development. While mature needles cannot be infected, the disease can be lethal and results in defoliation, making trees look unsightly. Conifer fungal diseases are slow to manage, and treatment focuses on protecting new growth. These trees still provide privacy even when partially defoliated. **Apple Scab** on crabapples is also a concern, but treatment can extend leaf retention for a month.

Name

Signature

CEU Sign Up Sheet

Topic: **Cabling Bracing**

Date: **9/22/24**

Meeting length: 60 Minutes

Meeting Summary:

Cabling and bracing are used to support high-value trees with structural issues that pruning alone cannot correct, particularly when located near valuable targets like homes or pools. Cables relieve torque from weak branch unions, while bracing rods help close cracks in the stem, promoting wound closure and faster compartmentalization. Steel cables enhance a tree's ability to generate wound response tissue. Bracing is crucial when wounds below branch unions can't be closed by cabling alone. Cottonwoods, due to their soft wood, are poor candidates for cabling. A case study on a cracked tree showed it required two bracing rods, two cables, and weight reduction pruning to improve structural integrity and prevent further damage.

Name

Signature

[illegible][illegible]

CEU Sign Up Sheet

Topic: **Native Oaks of Minnesota - Tree ID**

Date: **9/30/24**

Meeting length: 60 Minutes

Meeting Summary:

Discussed differences between and reviewed pictures of Red Oak (*Quercus rubra*), Northern Pin Oak, White Oak (*Quercus alba*), Eastern Pin Oak, Bur Oak (*Quercus macrocarpa*), and Swamp White Oak (*Quercus bicolor*)

The **Fagaceae** family includes **oaks**, **chestnuts**, and **beeches** (e.g., **Quercus**, **Castanea**, **Fagus**), distinguished by their cupule nuts, with some nut portions exposed. Typically, each nut contains a single seed, and these trees are characterized by high density wood and slow growth rates. They have **catkins** instead of flowers.

Red Oaks (e.g., **Quercus rubra**, **Q. palustris**) have acorns that mature in two years, with leaf lobes ending in bristles and smoother acorn edges. In contrast, **White Oaks** (e.g., **Q. alba**, **Q. macrocarpa**) have acorns that mature in one year, with rounded leaf lobes and rougher, hairier acorns.

Name

Signature

CEU Sign Up Sheet

Topic: **Pine Bark Adelaide Pineus Strobi**

Date: **10/6/24**

Meeting length: 60 Minutes

Meeting Summary:

Hemiptera includes aphids, adelgids, leafhoppers, and cicadas, all of which have piercing/sucking mouthparts, often referred to as "sap suckers." **Pine Bark Adelgids** typically infest **White Pine**, but can also affect **Scotch** and **Austrian pines**. While they're not lethal to mature trees, they can cause significant damage to smaller trees, especially when combined with other stress factors. They feed on phloem and secrete protective wax. Control methods include systemic insecticides and horticultural oil sprays.

Name

Signature

CEU Sign Up Sheet

Topic: **Pruning Mature Trees - Prof. Ed Gilman**

Date: **10/13/24**

Meeting length: 60 Minutes

Meeting Summary:

Professor Ed Gilman, a leading expert on tree pruning, recommends a careful, structured approach to pruning mature trees to ensure their health and structural integrity. His key advice includes removing dead, diseased, or damaged branches to reduce potential hazards. He emphasizes the importance of selective pruning, focusing on enhancing the tree's natural shape rather than over-pruning. Gilman advises pruning to improve air circulation and light penetration, which helps reduce disease risk. He suggests cutting branches back to lateral branches or the collar to encourage proper healing and minimize stress on the tree. Overall, the goal is to maintain the tree's overall health, structure, and longevity.

Name

Signature

CEU Sign Up Sheet

Topic: **Fall Diagnostics and Plant Health Care Treatment Options**

Date: **11/4/24**

Meeting length: 60 Minutes

Meeting Summary:

In fall, diagnosing tree health requires assessing changes since summer in both conifers and deciduous trees. For conifers, which retain needles, fungal damage and insect feeding are more evident, although older needles may have already fallen. Issues like European pine sawfly eggs can be visible, and needle shedding, particularly on white pines, is normal. Larch (Tamarack) sheds all its needles, unlike evergreens. Fall is also a time to consider sanitation and potential spring treatments.

For deciduous trees, leaf distortion makes diagnostics harder, and environmental factors like weather, disease, and soil fertility play a larger role. Fungal diseases may persist, but insects feeding on current leaves won't affect new growth. Fall watering is vital to prevent desiccation of root hairs, which are crucial for water uptake. This helps trees prepare for winter, while fall treatments for chlorosis, fungal issues, and poor soil can improve tree health going into the next growing season.

Name

Signature

CEU Sign Up Sheet

Topic: **Elms of MN Tree ID**

Date: **11/11/24**

Meeting length: 60 Minutes

Meeting Summary:

Reviewed similarities and differences between American, Siberian, Red, and Rock elm trees. Discussed bark, leaf, flower, and fruit characteristics, typical habitat, height, and fall foliage colors of each species.

Dutch Elm Disease (DED) is fatal to native elm species, with *Ulmus pumila* (Siberian Elm) being resistant but still susceptible to illness. Immediate treatment, including pruning diseased canopy parts, is crucial to prevent spread. Untreated trees can infect others, especially through grafted root systems. There's confusion around the term "Chinese Elm," often mistakenly applied to Siberian Elm. The true Chinese Elm, *Ulmus parvifolia* (Lacebark Elm), is highly resistant to DED, Elm Leaf Beetle, and Japanese Beetle. It has distinctive, exfoliating bark and is hardy in Zone 5, although it doesn't thrive in all climates.

Name

Signature

CEU Sign Up Sheet

Topic: Health and environmental impacts of urban forests

Date: **1/27/25**

Meeting length: 60 Minutes

Meeting Summary:

Green infrastructure, such as urban parks and forestry, provides numerous benefits for energy efficiency, health, and the environment. It reduces greenhouse energy demands for nearby buildings by 10% and enhances health by lowering heart rate, blood pressure, and heat-related illnesses while boosting the immune system. A 10% increase in tree cover can significantly reduce heat-related deaths, saving lives annually. Natural views align with human evolutionary preferences for safety and resources, while increased greenery near homes reduces non-accidental mortality. Schools with more tree cover see improved student performance, and urban forests offset carbon emissions from 10 million cars, with the potential to offset emissions from 15 million more through expanded tree planting. Research highlights these impacts, including a Portland initiative that prevented one premature death per 100 trees planted.

Name

Signature
